IT8204 Blockchain and Cryptocurrency - Week 6: Key Articles of Bahrain's Data Protection Law, and the Immutability Problem |

Marking rubric for your worksheet. You are scored 0-5 on each criterion (decimals allowed). Mark % = score x 20; your overall mark is the weighted average of the criteria, shown as a score out of 100. This worksheet contributes its weighted share to the A2 coursework total; the course result is decided on the overall course mark.

Score	Level	What it means		
0	**Missing**	No evidence / not attempted.		
1	**Limited**	Attempted but cannot give a basic explanation; mostly incorrect.		
2	**Basic**	Below expectation; major gaps.		
3	**Reasonable**	Meets the task; major enhancement still possible.		
4	**Very Good**	Minor enhancement only; valid and well justified.		
5	**Excellent**	Fully meets the task: every required part present, correct and specific.		

#	Criterion	Weight	What 'Excellent' (5) looks like	What 'Reasonable' (3) looks like	What 'Limited' or 'Basic' (1-2) looks like
C1	S1 T2 Individual build task Session 1 worksheet, Task 2: a Bahraini exchange wrote know your customer (KYC) data straight onto its chain. A customer now asks for erasure under Personal Data Protection Law (PDPL) Article 23.	**13%**	Gives ONE design change that would let the exchange meet the request. States honestly what the change still does not solve.	Gives a workable design change, but does not say what it still fails to solve, or the change only partly allows erasure.	No design change, or a change that would not allow erasure of the personal data.
C2	S2 T2 Individual build task Session 2 worksheet, Task 2: who, if anyone, is the data controller in a permissionless decentralised finance (DeFi) protocol.	**13%**	Identifies all three candidates - the front-end operator, the decentralised autonomous organisation (DAO) and the developers - and explains why the question is hard to settle.	Names the candidates, but one is missing, or states that the question is hard without explaining why.	A single answer with no reasoning, or no engagement with the candidates.
C3	S1 T3 Group-activity product (own write-up) Session 1 worksheet, Task 3: your own data design for a Bahraini retail-bank loyalty programme.	**13%**	States what goes on-chain and off-chain, and justifies an erasure mitigation for Article 23. Shows how access control stops data being re-linked to a person, and states the remaining risk honestly.	The design covers most of these four points, but one is missing or stated without a reason.	A general description with no on-chain and off-chain split, or no erasure mitigation.

C4	S2 T3 Group-activity product (own write-up) Session 2 worksheet, Task 3: your own tension map of Aave or Compound against PDPL Articles 12, 22, 23 and 24, plus your group proposal.	**13%**	Each of the four articles is linked to a specific feature of the product. The controller question is stated, and one unresolved tension is recorded honestly. Your group of four across roles A to D is proposed.	Most articles are mapped, but some links are general, or the unresolved tension is not stated.	Articles are listed without linking them to the product, or the map is missing.
C5	T1 Reading comprehension (both sessions) Task 1 in both worksheets. Session 1 asks why immutability clashes with Article 23, and about your chosen mitigation and its limit. Session 2 asks about the controller, the hardest article to satisfy, and the compliance officer's duty.	**16%**	Every question is answered in two or three sentences, in your own words. Each answer is correct and explains why, not only what.	All questions are answered and broadly correct. Some answers are one line, repeat the reading, or skip the reason.	Several questions are missing or wrong, or the answers are copied from the reading.
C6	T4 Analysis / evaluation at NQF L8 Across the week: weighing privacy rights against how blockchains work.	**14%**	Admits what a design cannot solve and says why. Weighs competing duties, such as privacy, record-keeping and ethics, and reaches a reasoned position.	Some judgement is shown, but the answers mostly describe, or reasons are stated without support.	Description only, or opinions given with no reasons.
C7	T5 Bahrain / GCC grounding Across the week: accurate use of Bahrain's PDPL.	**10%**	Uses the named PDPL articles accurately and applies them to the Bahraini exchange or bank in the task, rather than in general terms.	A correct Bahrain or GCC reference is made, but it could be removed without changing the answer.	The PDPL is misquoted, or another country's law is presented as Bahrain's.
C8	T6 Communication and academic conduct Both worksheets: how clear your writing is, the length of each answer, your sources, and submission.	**8%**	Both worksheets are submitted. Answers are clear, plain and close to the expected length for each task. Every source you used is listed in Task 4. Listing is what is marked; APA 7th style is advised, not marked.	Both worksheets are submitted and can be understood. Some answers are far too long or too short, or a source is missing.	A worksheet is missing, answers are hard to follow, or sources were used but not listed.

Worked exemplar - what a good answer contains

A good design says where the data lives, and what the design still does not solve, as the loyalty-programme worked example does. There, the personal data stays off the chain, only a hash goes on it, and the answer says honestly that the pattern of transactions stays visible on the chain.

This week's evidence is judged on reasoning and the individual written products from both sessions.

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Criterion	Weight %	Score (0-5)	Weighted	Comment (cite the screenshot / page)
S1 T2 Individual build task	13	5.0	13.0	Your Session 1 Task 2 answer gives one clear change: KYC moves to an access-controlled off-chain database with only a salted hash on-chain, and deleting the record and its salt breaks the link to the person. You weigh this against CBB retention rules and explain that the exchange can only block processing until the retention period ends. Your limits are specific: plaintext KYC in earlier blocks, and the hash and transaction pattern visible to nodes. To improve further, you could say how the permissioned members might migrate the ledger to remove the earlier blocks.
S2 T2 Individual build task	13	5.0	13.0	Your Session 2 Task 2 answer is complete. You name all three candidates with their roles: the front-end operator hosts the interface and sees IP addresses but cannot change records, the DAO votes on upgrades and risk rules but has no single legal identity, and developers published the code but do not run or approve transactions. Your reason, that each only partly qualifies and none controls the full life of on-chain data, is clear. To improve further, you could say which candidate a Bahraini regulator would most likely approach first, and why.

S1 T3 Group-activity product (own write-up)	13	5.0	13.0	Your loyalty-programme design covers all four parts in depth. You include a text sketch, keep CPR, name, phone and account details off-chain in Manama, and put only a salted hash of a random loyalty ID on-chain, explaining why a hashed CPR number could be guessed. You justify off-chain storage against crypto-shredding and state its cost. Access control works at two levels, with the lookup table and salts held under role-based access and logging. Your residual risk, timestamps and point values matched with till receipts, is specific. To improve further, you could shorten the write-up by removing repeated points.
S2 T3 Group-activity product (own write-up)	13	5.0	13.0	Your Aave tension map is detailed and specific. You link Article 12 to validator replication, Article 22 to the health factor and the liquidation function, Article 23 to permanent supply, borrow, repay and liquidation logs, and Article 24 to transaction signing that bundles execution with global publication. You state the controller question, document the Article 23 tension against AML record-keeping honestly, and propose four named people across roles A to D. To improve further, you could note the Article 22(2) exception for decisions taken in performing a contract.

T1 Reading comprehension (both sessions)	16	5.0	16.0	All six reading answers are correct and give a reason, and you develop them with care. Your Session 1 Q2 says when you would switch to a redactable ledger, your Session 1 Q3 explains re-identification through till receipts, and your Session 2 Q2 explains why Articles 22 and 24 can be partly handled while Article 23 cannot. Your Session 2 Q3 covers honest advice, a written record, escalation and refusing to sign off. To improve further, you could keep each answer closer to three short sentences, as the task suggests.
T4 Analysis / evaluation at NQF L8	14	5.0	14.0	Your judgement is a clear strength. You admit what each design cannot solve and why, including plaintext KYC in earlier blocks and permanent borrowing logs. You weigh competing duties directly, setting Article 23 privacy rights against CBB record-keeping and explaining when blocking replaces deletion. You reach a reasoned position: you would refuse to sign off raw permissionless Aave for retail customers, and you say what would change that view. To improve further, you could rank the risks you name by likelihood and impact.

T5 Bahrain / GCC grounding	10	5.0	10.0	Your use of the PDPL articles is accurate and applied. You give Article 23 with its ten working days and fifteen-day notice to third parties, Article 12 with the adequacy list, Article 22 as the right to object and ask for human reconsideration, and Article 24 as freely given, specific and revocable consent. You apply them to the exchange, the bank in Manama and a Bahraini Aave user. To improve further, take care with Decree No. 78 of 2019: it gives the Authority's powers to the Ministry of Justice for now, but it does not make any firm a controller.
T6 Communication and academic conduct	8	4.0	6.4	Both worksheets are submitted, your answers are clear and organised under labelled headings, and Task 4 lists your sources in full in both sheets, including the Law, the CBB Rulebook, the Finck report, the Aave and Etherscan pages and the course materials. However, your Session 2 Task 3 write-up runs to about 690 words, well beyond the half page (about 250 words) the task expects, so it is longer than a reader needs. To improve further, choose the two or three strongest points for each article and cut the rest.
TOTAL	**100**		**98.4**	*Weights sum to 100; max weighted = 100.*
Overall mark (out of 100)	**98.4**	*Weighted average of the criteria, out of 100. This worksheet contributes its weighted share to the A2*		

Overall feedback to the student

This is a very strong Week 6 submission. Your answers are accurate, detailed and well reasoned. In Session 1 you moved KYC off-chain with a salted hash, weighed erasure against CBB record-keeping, and named the plaintext left in earlier blocks. Your loyalty design explains why a random loyalty ID is safer than a hashed CPR number and controls access at two levels. Your Aave map links every article to real contract functions, such as the liquidation call. The one mark held back is length: your Session 2 Task 3 write-up is far longer than the half page asked. As the group business case begins in Week 7, practise keeping each section within its expected length by stating the point, the evidence and the decision once.